Tech Trends: Gate Drivers Comparison

	Microchip Technology Gate Drivers	Texas Instruments DRV8706-Q1	Infineon Technologies EiceDRIVERs	Texas Instruments DRV8300	Texas Instruments UCC5870-Q1	ON Semi-conductor NCP51810	Texas Instruments UCC27284-Q1	Texas Instruments UCC21710	Trinamic TMC6100 Power Driver	Texas Instruments UCC23514	STMicro-electronics MASTERGAN4	Silicon Labs Si828x drivers
Designed to drive	MOSFETs	High-side and low-side N-channel power MOSFETs	MOSFETs, IGBTs, and SiC MOSFETs	High-side and low-side N-channel power MOSFETs	High power SiC MOSFETs and IGBTs	GaN power switches	N-channel MOSFETs	1700V SiC MOSFETs and IGBTs	MOSFETs	IGBTs, MOSFETs, and SiC MOSFETs	GaN power transistors	Silicon Carbide (SiC) FETs, IGBTs, and silicon MOSFETs
Input voltage range	4.5V-18V	4.9V-37V		4.8V-20V	3V-5.5V	Up to 20V	5.5V-16V	5V	8V-60V	Up to 35V	3.3V to 15V	30V
Typical propaga-tion delay	27ns	Maximum 850ns	30ns	125ns	150ns maximum	50ns maximum	16ns	130ns	-	105ns maximum	70ns	50ns maximum
Isolation	No	No	Yes	No	Yes	Yes	No	Yes	No	Yes	No	Yes
CMTI	-	-	Highest CMTI >200kV/µs		100kV/µs minimum	-	-	>150V/ns	-	150-kV/µs minimum	-	125kV/µs
Protection features	Latch-up resistant, protected against ESD, accept upto 500mA reverse current	1. Dedicated driver disable pin (DRVOFF) 2. Supply and regulator volt-age monitors 3. MOSFET VDS overcurrent monitors 4. MOSFET VGS gate fault monitors 5. Charge pump for reverse po-larity MOSFET 6. Offline open load and short circuit diagnostics 7. Device thermal warning and shutdown 8. Fault condi-tion interrupt pin (nFAULT)	DESAT, soft-Off, UVLO, miller clamp, Two Level Turn Off (TLTO), and fault function	1. BST under-voltage lockout (BSTUV) 2. GVDD undervoltage (GVDDUV)	Miller clamp, fault alarm, high voltage clamping control, under-voltage and over-voltage protection	Under-voltage lockout (UVLO), monitoring VDD bias voltage, and VDDH and VDDL driver bias, and thermal shutdown	UVLO, over-voltage and overcurrent protection	Fast over-current and short circuit detection, shunt cur-rent sensing support, fault report-ing, active Miller clamp, and input and output side power supply UVLO	Full protec-tion and diagnostics via SPI interface	Overvoltage, overcurrent protection, 13V reverse polarity volt-age handling capability on the input stage and miller clamp	UVLO protec-tion, prevent-ing the power switches from operating in low efficiency or dangerous conditions, and the interlocking function	Miller Calmp, DESAT detection, FAULT feedback, UVLO, Soft turn-off
Interface	-	1. SPI: Detailed configuration and diagnostics 2. H/W: Simpli-fied control and fewer MCU pins	I2C	-	SPI based device recon-figuration	-	-	-	SPI	-	Easy interfac-ing with mi-crocontrollers, DSP units or Hall effect sensors.	-
Package	8-lead MSOP, 8-lead SOIC, 8-lead 2 x 3 TDFN	Compact VQFN package	Space-saving DS0-16 fine pitch wide-body package with 8mm creepage	Compact QFN and TSSOP packages	12.8 mm×7.5 mm SSOP (36) package	QFN15 package	SOIC packages	DW SOIC-16 package	7mm× 7mm QFN package	DWV package with 8.5mm creepage	9mm×9mm QFN package	16-pin, 20-pin, and 24-pin Wide-Body SOIC
Operating teperature	-40°C to +125°C	-40°C to +125°C	Upto 125°C	-40°C to +125°C	-40°C to +150°C	Upto 150°C	–40°C to 150°C	–40°C to 150°C	-40°C to +125°C	–40°C to +150°C	-40°C to 125°C	-40 to 125°C
Applica-tions	1. Switch-mode power supplies 2. Pulse transformer drive 3. Line drivers 4. Level translator 5. Motor and solenoid drive	1. Automotive brushed DC motors 2. Solenoids and relays 3. Power window lift and sliding door 4. Power sunroof 5. Power seat modules 6. Power trunk and lift gate 7. BDC fuel, water, oil pumps 8. Windshield wipers	1. Solar string inverters 2. EV charging 3. UPS 4. Industrial drives 5. Common Access Card (CAC) 6. Welding equipment 7. Induction heating appliances 8. Power supplies for server and telecom-munication systems	1. E-Bikes, e-scooters, and e-mobility 2. Fans, pumps, and servo drives 3. Brushless-DC (BLDC) motor modules and PMSM 4. Cordless garden and power tools, lawnmowers 5. Cordless vacuum cleaners 6. Drones, robotics, and RC toys 7. Industrial and logistics robots	1. High-power IGBTs and SiC MOSFETs 2. HEV and EV traction inverter 3. HEV and EV power modules	1. Full or half-bridge 2. Active clamp flyback or forward and synchro-nous rectifier topologies 3. 48V to 12V PoL converters 4. 48V to low voltage bus converter 5. Industrial modules	1. Automotive DC/DC converters 2. Electric power steering 3. Onboard charger (OBC) 4. Integrated belt starter generator (iBSG) 5. Automo-tive HVAC compressor modules	1. Industrial motor drives 2. Server, telecom, and industrial power supplies 3. Uninter-ruptible power supplies (UPS) 4. Solar inverters 5. Renew-able energy storage converters	1. PMSM FOC drives and BLDC motors 2. Industrial drives 3. Factory automation 4. Lab automation 5. Robotics 6. CNC machines 7. Textile machines 8. Pumps 9. Survei-llance cameras 10. Home automation 11. Printers	1. Industrial motor-control drives 2. Industrial power sup-plies, UPS 3. Solar inverters 4. Induction heating	1. Switch-mode power supplies 2. Chargers and adapters 3. High-voltage PFC, DC-DC and DC-AC Converters	In a wide variety of inverter and electric vehicle systems